

Amir Reza Isazadeh

PHD IN REHABILITATION SCIENCE

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Synopsis

Researcher and R&D Engineer specializing in **tissue biomechanics and simulations**, with a record of turning complex models into **clinically viable software and medical devices**.

Professional Profile

Researcher at Head and Neck Simulation Lab

Alberta, Canada

UNIVERSITY OF ALBERTA

2023-2025

- Built a **first-of-its-kind**, patient-specific **Virtual Surgical Planning (VSP)** pipeline for free-flap reconstruction.
- **100× Performance Boost**: Developed a **real-time solver** that outperforms FEM-based optimization by two orders of magnitude and ensures clinically viable surgical designs (open-source).
- Engineered a robust, **fully automated, end-to-end surgical simulation pipeline** for large-scale surgical outcome prediction and individualized planning (open-source).
- Reduced **MRI-to-simulation** preprocessing time from ~4 hours to **~15 minutes** using **AI-powered image segmentation**, enabling same-day individualized surgical planning.
- Designed an additively manufactured **surgical guiding toolkit** that delivers **millimeter-level accuracy** when translating VSPs into surgical execution.
- Accelerated **clinical adoption** by partnering with surgeons and experts at the UofA Hospital and iRSM (Misericordia Hospital), completing 40+ hours of OR shadowing, and iterating the VSP pipeline based on surgeons' feedback.
- Secured a total of **\$270K** in competitive scholarships and research funding through strategic proposals and earned a nomination for the prestigious **Vanier Scholarship**.
- Completed four impactful studies ahead of schedule, yielding a **PhD in under three years**.
- Served as a **reviewer for IEEE** and **mentored junior researchers** across diverse projects.

Research Assistant

Tehran, Iran

CENTER OF EXCELLENCE IN DESIGN, ROBOTICS, AND AUTOMATION

2021

- Developed a **rapid optimization framework** for automotive manufacturing process design; published the methodology in high-impact journals.
- **Led a team of 4** to pilot the workflow with IKCO (automotive manufacturing); integration assessment projected $\approx 35\%$ reduction in defective parts by improving upfront process design decisions.

Educational Excellence

PhD in Rehabilitation Science

Alberta, Canada

UNIVERSITY OF ALBERTA

2023–2025

Thesis: Patient-Specific Virtual Surgical Planning for Tongue Reconstruction: Biomechanical Optimization, Surgical Simulation, and Guided Execution **GPA: 3.95/4.0**

M.Sc. in Mechanical Engineering

Tehran, Iran

SHARIF UNIVERSITY OF TECHNOLOGY

2019–2021

Thesis: Development of a Rapid Optimization Method and Its Application in Sheet Metal Forming Process Design **GPA: 4.0/4.0**

B.Sc. in Mechanical Engineering

Tehran, Iran

AMIRKABIR UNIVERSITY OF TECHNOLOGY

2013–2018

Thesis: Algorithmic Design of Axial-Flow Turbomachinery Compressors **GPA: 3.9/4.0**

Publications

1. **Isazadeh A**, Chen S, Seikaly H, Zheng B, Westover L, Aalto D (**2025**) Guided tongue reconstruction: Feasibility analysis of additively manufactured surgical toolkit on a silicone tongue model. *Surgical Oncology*. (under review).
2. **Isazadeh A**, Westover L, Seikaly H, Aalto D (**2025**) Computational analysis of tongue reconstruction surgery: The impact of free flap site and volume on the tongue anatomy and biomechanics. *PLOS ONE*. (under review).
3. **Isazadeh A**, Zenke J, Westover L, Seikaly H, Aalto D (**2025**) Patient-specific virtual surgical planning for tongue reconstruction: Evaluating hyperelastic inverse FEM with four simulated tongue cancer cases. *Phys Med Biol* 70:145017. doi: 10.1088/1361-6560/adebd8
4. **Isazadeh A** and Aalto D (**2025**) VSP-BioSim: Patient-Specific Virtual Surgical Planning via Biomechanical Simulations. *Open-source*. doi: 10.5281/zenodo.17438091
5. **Isazadeh A** and Aalto D (**2025**) hiFEM: Hyperelastic Inverse Finite Element Method. *Open-source*. doi: 10.5281/zenodo.17427896
6. **Isazadeh A**, Seikaly H, Westover L, Aalto D (**2024**) Algorithmically designed flaps in tongue reconstruction: A feasibility analysis. *Int J Comput Assist Radiol Surg* 19, 735–746. doi: 10.1007/s11548-024-03062-w
7. Shamloofard M, **Isazadeh A**, Shirin MB, Assempour A (**2022**) Optimum design of middle-stage tool geometry and addendum surfaces in sheet metal stamping processes using a new isogeometric-based framework. *Proc Inst Mech Eng Part B J Eng Manuf* 236:741–757. doi: 10.1177/09544054211041046
8. **Isazadeh A**, Shamloofard M, Assempour A (**2021**) Some improvements on the one-step inverse isogeometric analysis by proposing a multi-step inverse isogeometric methodology in sheet metal stamping processes. *Applied Mathematical Modeling* 91:476–492. doi: 10.1016/j.apm.2020.09.032

Conference Presentations

- Virtual planning of tongue cancer surgery: patient-specific simulation of hemiglossectomy biomechanics. *19th International Symposium on Computer Methods in Biomechanics and Biomedical Engineering (CMBBE)*. Vancouver, Canada. 2024.
- Algorithmically designed flaps in tongue reconstruction: A feasibility analysis. *Prairie Collaborative Research Conference*. Edmonton, Canada. 2023.
- Analysis of a sheet metal stamping process with high drawing depth using a new NURBS-based methodology. *28th International Conference of Mechanical Engineers Association of Iran*. Tehran, Iran. 2020.

Recognition, Awards & Scholarships

- Graduate Student Support Award (\$5,000) Canada
GPS Awards, University of Alberta 2024
- Nominated for Vanier Canada Graduate Scholarships Canada
CIHR Federal Competition 2023
- Mickleborough Scholarship in Rehabilitation Science (\$1,000) Canada
University of Alberta 2023
- Faculty of Rehabilitation Medicine GRAF (\$64,000) Canada
University of Alberta 2023
- Graduate Recruitment Scholarship (\$5,000) Canada
University of Alberta 2023
- Industrial Transformation and Training Centre for Joint Biomechanics Scholarship (\$99,000) Australia
Queensland University of Technology (successful, declined to pursue PhD at UofA) 2022
- Higher Degree Research Tuition Fee Sponsorship (\$96,000) Australia
Queensland University of Technology (successful, declined to pursue PhD at UofA) 2022
- Recognized as “Outstanding Student” and ranked in the top 5% of graduates, Mechanical Engineering Department Iran
Sharif University of Technology 2021
- Ranked 11th (out of 13,900) in the Iranian national M.Sc. entrance exam Iran
National Organization of Educational Testing 2018
- Qualified for double B.Sc. degree program, Exceptional Talents Office Iran
Amirkabir University of Technology 2015
- Tuition waiver scholarship for B.Sc. program Iran
National Organization of Educational Testing 2015
- Ranked in the top 1% in the Iranian national B.Sc. entrance exam Iran
National Organization of Educational Testing 2013

Technical Mastery

COMPUTATIONAL MECHANICS & SIMULATION

- Nonlinear Finite Element Method (FEM), inverse FEM
- Soft Tissue Biomechanics, Large Deformations, Visco-hyperelastic Modeling
- Real-time Simulations; NVIDIA Warp, Omniverse, Isaac Sim

VIRTUAL SURGICAL PLANNING

- AI-Driven MRI Segmentation
- Automated Patient-Specific Model Preprocessing Pipeline

TOOLS

- *FEM solvers* : FEBio, ABAQUS, PolyFEM, NVIDIA Warp
- *Mesh Generation* : fTetWild, Interactive All-hex Meshing, HexBox, and PolyCube libraries
- *Geometric Processing* : Open3D, PyVista
- *Geometric Modeling* : SolidWorks, MeshMixer, Rhino/Grasshopper, Autodesk Fusion360, Autodesk Inventor
- *Programming* : Python, MATLAB
- *AI/ML* : Scikit-learn, PyTorch
- *Optimization* : Genetic Algorithms, Neural Networks
- *Medical Image Segmentation* : Slicer 3D, Biomedisa (AI-powered)
- *Statistics* : statsmodels, R, Seaborn, STATA
- *Data Visualization* : Seaborn, Matplotlib, Paraview
- *Version Control* : Git, GitHub
- Attended numerous workshops such as Biomedisa, ArtiSynth, FEBio, etc.

TEACHING & OUTREACH

- Volunteered for MRI pilot studies—University of Alberta 2023
- Teaching Assistant—Metal Forming Processes and Strength of Materials 2019-2020
- Co-founded "Konkursite," a non-profit organization that provided online education to over 8,000 underserved students 2017–2020